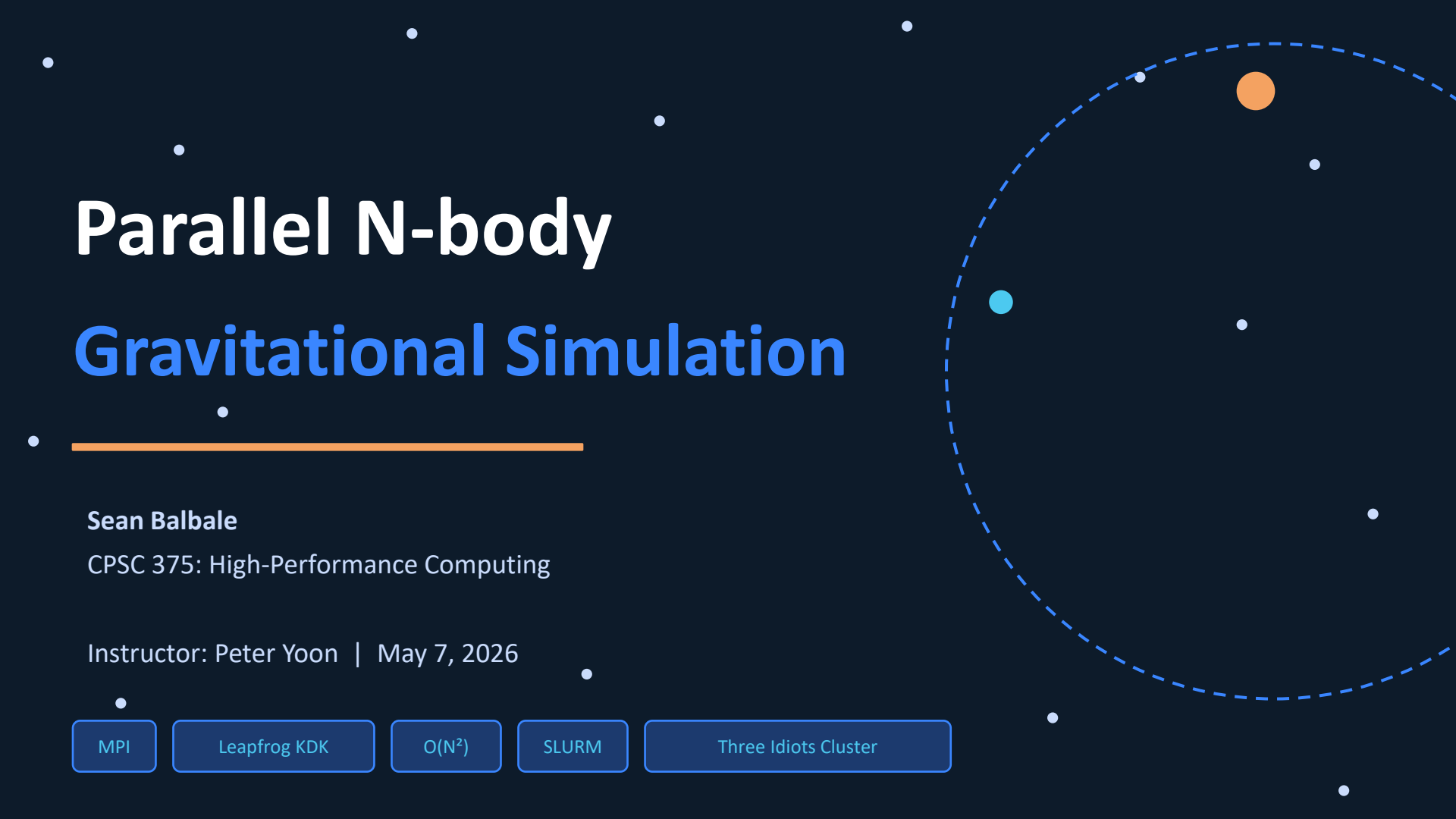


Parallel N-body

Gravitational Simulation



Sean Balbale

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Instructor: Peter Yoon | May 7, 2026

MPI

Leapfrog KDK

$O(N^2)$

SLURM

Three Idiots Cluster

N-body Gravitational Simulation

The Problem

Simulate N particles moving under mutual gravitational attraction — each particle exerts force on every other. Classic in astrophysics: galaxy dynamics, planetary systems, dark matter halos.

Computational Challenge

Naïve all-pairs force evaluation is $O(N^2)$ per timestep. At $N = 9,600$ and 10 steps, a serial run takes ~ 1 second. Scaling to millions of particles requires parallelism.

Why MPI?

No shared memory across cluster nodes. Distributed-memory parallelism via MPI lets us exploit all 3 nodes and 24 cores of the Three Idiots cluster simultaneously.

Softened Force Law

$$\mathbf{F}_{ij} = \mathbf{G} \cdot \mathbf{m}_j \cdot (\mathbf{r}_j - \mathbf{r}_i) \\ (|\mathbf{r}_j - \mathbf{r}_i|^2 + \epsilon^2)^{3/2}$$

$\epsilon = 0.05$ (Plummer softening)

$G = 1.0$ (N-body units)

9,600

Max particles

$O(N^2)$

Per timestep

3 nodes

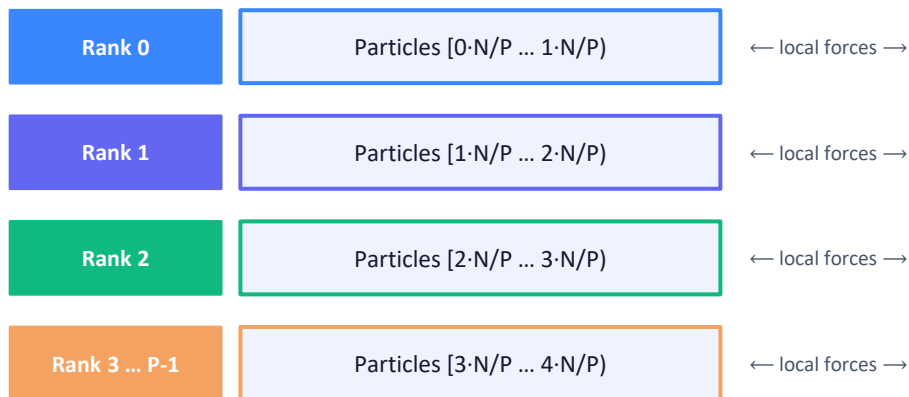
24 cores total

Algorithm & Parallel Design

Leapfrog (KDK) Integration

- 1 **Half-kick** $v += (F/m) \cdot \Delta t / 2$
- 2 **Drift** $x += v \cdot \Delta t$
- 3 **Allgather $\times 3$** Sync all positions across P ranks
- 4 **Force eval** $F = G \cdot \sum m_j \cdot (r_j - r_i) / (r^2 + \epsilon^2)^{3/2}$
- 5 **Half-kick** $v += (F/m) \cdot \Delta t / 2$

MPI Decomposition



MPI_Allgather $\times 3$ (x, y, z positions)

3 collective calls per timestep · O(N) data per call · $\Delta t = 0.01$

Cluster Configuration & Study Design

Cluster Name

Three Idiots

Nodes

3 nodes

Total Cores

24 MPI procs

Fabric

TCP/IP 10.0.0.0/24

Scheduler

SLURM + PMI2

Compiler

Intel mpiicx -O3 -xHost

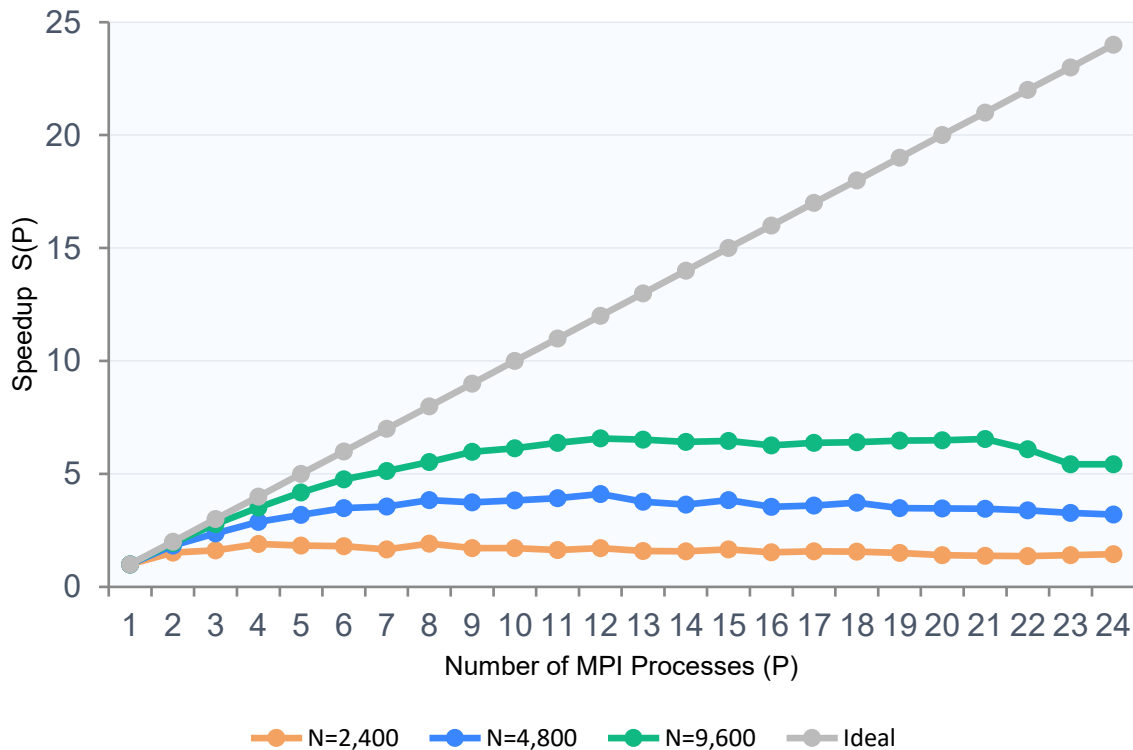
Scaling Studies

Study	Particles (N)	Steps	P range
Strong	2,400	10	1 – 24
Strong	4,800	10	1 – 24
Strong	9,600	10	1 – 24
Weak	1,000 × P	50	1 – 24

Methodology

3 trials per configuration · median wall time reported
MPI_Wtime() brackets simulation loop only
Seed 42 for reproducibility

Speedup $S(P) = T(1) / T(P)$



N = 2,400 · P=8

Peak: 1.92x

N^2/P compute swamped
by Allgather overhead

N = 4,800 · P=12

Peak: 4.11x

Best speedup; comm still
manageable fraction

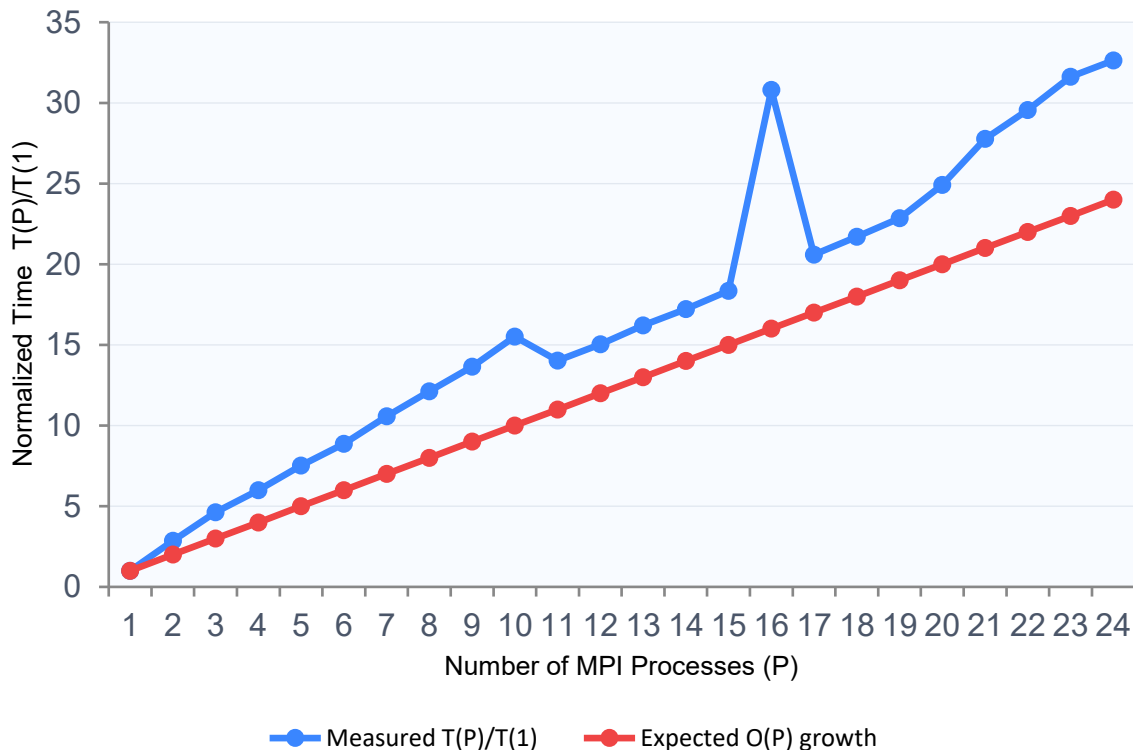
N = 9,600 · P=12

Peak: 6.57x

Largest N — best strong
scaling efficiency

All curves flatten at $P > 12$ — communication
overhead dominates

1,000 Particles Per Process · 50 Timesteps



$O(N^2)$ Complexity

Even perfect weak scaling can't give constant time: each process computes $O(N/P \times N) = O(N)$ forces as N grows. Normalized time grows as $O(P)$.

P=16 Anomaly (1.70s)

Sharp spike at $P=16$ aligns with crossing the 2-node TCP boundary. Buffer saturation on the inter-node fabric inflates Allgather latency $\sim 2\times$.

P=11 Dip (0.77s)

Temporary drop as SLURM packs rank assignment within one socket, reducing NUMA latency — disappears at $P=12$ when ranks spill across cores.

Extracting the Serial Bottleneck

$$S(P) = \frac{1}{f + (1 - f) / P}$$

f = serial fraction

Problem Size	Serial Fraction f	Theoretical Max 1/f	Actual Max (P=24)
N = 2,400	68%	≈ 1.5×	1.92×
N = 4,800	28%	≈ 3.6×	4.11×
N = 9,600	15%	≈ 6.7×	6.57×

Key insight: larger N ↓ serial fraction — Allgather overhead is fixed cost amortized over more compute work

Conclusions & Takeaways

SS

Strong Scaling

Speedup plateaus at $P \approx 12$ for all N . Larger problem sizes achieve better peak speedup: $6.57\times$ at $N=9,600$ vs. only $1.92\times$ at $N=2,400$.

WS

Weak Scaling

Normalized time grows super-linearly with P due to $O(N^2)$ complexity. Three Allgather calls per step scale as $O(P)$ communication overhead.

f

Amdahl's Law

Serial fraction drops from 68% to 15% as N quadruples — the $3\times$ Allgather fixed cost is amortized over more $O(N^2)$ compute.

!

P=16 Anomaly

Sharp $2\times$ spike in weak scaling at $P=16$ — attributed to inter-node TCP buffer saturation when ranks span all three nodes.

Future: replace $3\times$ Allgather with ring-pass ghost exchange $\rightarrow O(N/P)$ comm instead of $O(N)$